



**CALICUT UNIVERSITY – FOUR-YEAR UNDER  
GRADUATE PROGRAMME (CU-FYUGP)**

**BSc CHEMISTRY**

|                |                                                                                                                                                                           |                  |                   |                    |             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | B. Sc. Chemistry                                                                                                                                                          |                  |                   |                    |             |
| Course Title   | <b>ENVIRONMENTAL CHEMISTRY</b>                                                                                                                                            |                  |                   |                    |             |
| Type of Course | <b>MDC</b>                                                                                                                                                                |                  |                   |                    |             |
| Semester       | <b>I</b>                                                                                                                                                                  |                  |                   |                    |             |
| Academic Level | <b>100-199</b>                                                                                                                                                            |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                    | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 3                                                                                                                                                                         | 3                | -                 | -                  | 45          |
| Pre-requisites | What is Environment.<br>Basic idea of environmental pollution.                                                                                                            |                  |                   |                    |             |
| Course Summary | This course ensures that the students acquire a profound knowledge and understanding on environmental pollution and the necessity of controlling environmental pollution. |                  |                   |                    |             |

**Course Outcomes (CO):**

| CO  | CO Statement                                                                                                  | Cognitive Level* | Knowledge Category# | Evaluation Tools used                          |
|-----|---------------------------------------------------------------------------------------------------------------|------------------|---------------------|------------------------------------------------|
| CO1 | Acquire the knowledge on ecosystem.                                                                           | U                | C                   | Instructor-created exams / Quiz                |
| CO2 | Recall the technical/scientific terms involved in pollution.                                                  | U                | C                   | Instructor-created exams / Quiz                |
| CO3 | Recognize different types of toxic substances that cause environmental pollution.                             | U                | C                   | Instructor-created exams / Assignment          |
| CO4 | Understand the effects of environmental pollution.                                                            | U                | C                   | Seminar Presentation / Viva                    |
| CO5 | Understand various pollution control measures.                                                                | U                | C                   | Instructor-created exams / Quiz                |
| CO6 | Discuss and report local and global environmental issues based on the knowledge gained throughout the course. | Ap               | P                   | Group discussion and Seminar presentation/Viva |

\* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)  
 # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)  
 Metacognitive Knowledge (M)

**Detailed Syllabus:**

| Module     | Unit                                           | Content                                                                                                                                                                          | Hrs       | Mark      |
|------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| <b>I</b>   | <b>Introduction to Environmental Chemistry</b> |                                                                                                                                                                                  | <b>9</b>  | <b>18</b> |
|            | 1                                              | Environmental segments-Atmosphere, Hydrosphere, Lithosphere, Biosphere                                                                                                           | 2         |           |
|            | 2                                              | Interaction between different environmental spheres Concept of ecosystem, abiotic and biotic components                                                                          | 2         |           |
|            | 3                                              | Composition of Air, Water and Soil                                                                                                                                               | 2         |           |
|            | 4                                              | Environmental pollution – Concepts and definition – Pollutant, contaminant, receptor and sink                                                                                    | 1         |           |
|            | 5                                              | Classification of pollutants – Global, regional, local, persistent and non-persistent pollutants.                                                                                | 1         |           |
|            | 6                                              | Types of pollution                                                                                                                                                               | 1         |           |
| <b>II</b>  | <b>Air Pollution</b>                           |                                                                                                                                                                                  | <b>9</b>  | <b>18</b> |
|            | 7                                              | Tropospheric pollution – Gaseous air pollutants – Hydrocarbons, oxides of sulphur, nitrogen and carbon (Elementary idea only)                                                    | 2         |           |
|            | 8                                              | Global warming, green house effect, acid rain                                                                                                                                    | 1         |           |
|            | 9                                              | Particulates – Smog: London smog and photochemical smog –                                                                                                                        | 2         |           |
|            | 10                                             | stratospheric pollution - depletion of ozone layer, chlorofluorocarbons - Automobile pollution.                                                                                  | 2         |           |
|            | 11                                             | Control of air pollution                                                                                                                                                         | 2         |           |
| <b>III</b> | <b>Water Pollution</b>                         |                                                                                                                                                                                  | <b>10</b> | <b>20</b> |
|            | 12                                             | Impurities in water – cause of pollution – natural and anthropogenic – Marine water pollution – Underground water pollution.                                                     | 1         |           |
|            | 13                                             | Source of water pollution – Industrial waste, Municipal waste, Agricultural waste, Radioactive waste, Petroleum, Pharmaceutical, heavy metal, pesticides, soaps and detergents.  | 2         |           |
|            | 14                                             | Types of water pollutants: Biological agents, physical agents and chemical agents – Eutrophication- biomagnification and bioaccumulation.                                        | 2         |           |
|            | 15                                             | Water quality parameters: DO, BOD, COD, alkalinity, hardness, chloride, fluoride and nitrate. Toxic metals in water and their effects: Cadmium, lead and oil pollution in water. | 3         |           |
|            | 16                                             | Water pollution control methods                                                                                                                                                  | 2         |           |

| IV | Soil, Thermal, and Radioactive Pollutions |                                                                                                                                                                                                                                                             | 8 | 14 |
|----|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|
|    | 18                                        | Soil pollution: Sources by industrial and urban wastes. Non-degradable, degradable and biodegradable wastes. Hazardous waste.                                                                                                                               | 2 |    |
|    | 19                                        | Pollution due to plastics, pesticides, biomedical waste and <i>e-waste</i> (source, effects and control measures) – Control of soil pollution - Solid waste Management – Open dumping, Landfilling, Incineration, Re-use, reclamation, recycle, composting. | 3 |    |
|    | 20                                        | Thermal pollution – definition, sources, harmful effects and prevention.                                                                                                                                                                                    | 1 |    |
|    | 21                                        | Radioactive pollution (source, effects and control measures) – Hiroshima, Nagasaki and Chernobyl accidents (brief study).                                                                                                                                   | 2 |    |
| V  | Open Ended Module: Environmental issues   |                                                                                                                                                                                                                                                             | 9 |    |
|    | 1                                         | Environment and society<br><br>Pollution case studies: Chernobyl disaster, Bhopal tragedy, Endosulfan disaster in Kerala (brief study) etc.                                                                                                                 |   |    |

## References

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2. A. K. Ahluwalia, *Environmental Chemistry*, The Energy and Resources Institute, 2017.
3. Balram Pani, *Textbook of Environmental Chemistry*, I. K. International Pvt Ltd, 2010.
4. S.K. Banergy, *Environmental Chemistry*, 2<sup>nd</sup> Edn., Prentice-Hall of India Pvt. Ltd., New Delhi, 2005.
5. V.N. Bashkin, *Environmental Chemistry: Asian Lessons*, Springer Science & Business Media, 2003.
6. S.E. Manahan, *Environmental Chemistry*, 8<sup>th</sup> Edn., CRC Press, Florida, 2004.
7. Balram Pani, *Textbook of Environmental Chemistry*, I. K. International Pvt Ltd, 2010.
8. J. M. H. Selendy, *Water and Sanitation-Related Diseases and the Changing Environment*, John Wiley & Sons, Inc.
9. P. K. Goel, *Water Pollution: Causes, Effects and Control*, New Age International, 2006.
10. V. N. Bashkin, *Environmental Chemistry: Asian Lessons*, Springer Science & Business Media, 2003.
11. Pallavi Saxena, Vaishali Naik, *Air Pollution: Sources, Impacts and Controls*, CAB International, 2018.
12. Gabi Mocatta (2015) *Environmental Journalism*, Deakin University Open School of Journalism.
13. D. S. Poornananda (2022), *Environmental Journalism: Reporting on Environmental Concerns and Climate Change in India*, SAGE Publishing India
14. Frome, Michael. (1998) *Green Ink: An Introduction to Environmental Journalism*. Salt Lake City: University of Utah Press.
15. Sachsman, D. B. & Valenti, J. M. (2020). *Routledge handbook of environmental journalism*. New York, NY: Routledge.
16. Blum, D., Henig, R., Knudson, M., (2005). "[A Field Guide for Science Writers](#)." Oxford University Press; 2nd edition.
17. Hansen, Anders. (2010) *Environment, Media and Communication*. London: Routledge

**Mapping of COs with PSOs and POs :**

|         | PSO<br>1 | PSO<br>2 | PSO<br>3 | PSO4 | PS<br>O5 | PSO<br>6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---------|----------|----------|----------|------|----------|----------|-----|-----|-----|-----|-----|-----|-----|
| CO<br>1 | 1        | -        | -        | -    | 1        | 1        | 1   |     |     | 2   | 1   |     |     |
| CO<br>2 | 1        |          | -        | -    | 1        | 1        | 1   |     |     | 1   | 1   | 1   | 1   |
| CO<br>3 | -        | -        |          | 1    | 2        | 2        | 1   |     |     | 2   | 2   | 1   |     |
| CO<br>4 | -        | -        |          |      | 1        | 2        | 1   |     |     | 1   | 1   | 1   | 1   |
| CO<br>5 | -        |          | -        | 1    | 2        | 2        | 1   |     |     | 1   |     | 1   | 1   |
| CO<br>6 | -        | -        | -        | 1    | 2        | 2        | 1   |     |     | 1   | 1   | 1   | 1   |

**Correlation Levels:**

| Level | Correlation           |
|-------|-----------------------|
| -     | Nil                   |
| 1     | Slightly / Low        |
| 2     | Moderate /<br>Medium  |
| 3     | Substantial /<br>High |

**Assessment Rubrics:**

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics :**

|      | Internal Exam | Assignment/viva | Quiz/seminar/<br>Group discussion | End Semester Examinations |
|------|---------------|-----------------|-----------------------------------|---------------------------|
| CO 1 | ✓             |                 | ✓                                 | ✓                         |
| CO 2 | ✓             |                 | ✓                                 | ✓                         |
| CO 3 | ✓             | ✓               |                                   | ✓                         |
| CO 4 |               | ✓               | ✓                                 | ✓                         |
| CO 5 | ✓             |                 | ✓                                 | ✓                         |
| CO 6 |               | ✓               | ✓                                 |                           |